

Credit Risk Scoring Model

Credit Underwriting / Risk Management Portfolio Project | 500 Synthetic Borrowers

500 Borrower Sample	A-E Risk Tiers	PD x EAD x LGD Loss Framework	45% Illustrative LGD
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Model Overview

This project builds a transparent consumer-credit scorecard that evaluates borrower-level default risk and aggregates the results into a portfolio credit dashboard. The model converts key underwriting characteristics into risk points, an estimated probability of default (PD), an A-E risk tier, an underwriting decision and expected credit loss.

Credit Scorecard

Risk Factor	Low Risk	Moderate Risk	High Risk
FICO Score	≥ 740	670-739	< 670
Debt-to-Income	$\leq 25\%$	25-40%	$> 40\%$
Credit Utilization	$\leq 30\%$	30-60%	$> 60\%$
Delinquencies	0	1	2+
Employment History	≥ 5 yrs	2-5 yrs	< 2 yrs
Loan / Income	$\leq 25\%$	25-50%	$> 50\%$

Risk Segmentation & Underwriting

Tier	Risk Points	Indicative PD	Underwriting View
A	0-19	$< 3\%$	Low risk / strongest profile
B	20-39	3-7%	Acceptable
C	40-59	7-15%	Conditional / tighter structure
D	60-79	15-30%	Enhanced review
E	80+	$> 30\%$	Very high risk

Core Credit Analytics

- Probability of Default:** risk points are transformed into an estimated PD so borrowers can be compared on a common probability scale.
- Expected Loss:** the model applies Expected Loss = PD x Exposure at Default x Loss Given Default, using a 45% illustrative LGD.
- Portfolio monitoring:** the dashboard tracks total exposure, average PD, expected loss, default rate and exposure concentrated in D/E risk tiers.
- Validation:** borrowers with lower model credit scores are compared with higher-score borrowers to test whether realized synthetic defaults move in the expected direction.

Key Underwriting Insight

The model does not rely on a single credit metric. A borrower with an acceptable FICO score can still migrate into a weaker risk tier when high leverage, utilization, delinquencies, unstable employment or an oversized loan increase expected default risk. This demonstrates the difference between a credit score and a broader underwriting decision.

Recruiting / Interview Takeaway

I built a borrower-level credit risk scorecard and portfolio dashboard using FICO, DTI, utilization, delinquency history, employment stability and loan-to-income. I mapped those factors into risk points and PD estimates, calculated expected loss using PD x EAD x LGD, segmented borrowers into A-E risk tiers and added validation tests to evaluate whether the score separates higher- and lower-risk borrowers.

Methodology & Disclaimer

All 500 borrower records are synthetic and contain no real customer or personally identifiable information. The scorecard and PD calibration are illustrative and intended only for educational and recruiting use. A production lending model would require representative historical performance data, independent validation, model-risk governance, fair-lending and disparate-impact testing, adverse-action reason codes, drift monitoring and compliance review.